

CHE 210: HW #10

Lila Aryadwita

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- ① a) $\Delta \hat{H}^{\circ}_r = +69.36$, so we know the reaction is endothermic at 25°C which means it will absorb heat. Therefore, heat will be needed to keep the temperature constant. If the reaction ran adiabatically, the temperature would decrease because no heat transfer would occur. The energy required to break the molecular bonds is greater than the energy released when the bonds form based from $\Delta \hat{H}^{\circ}_r$.

b) $\Delta \hat{H}^{\circ}_r = \Delta(\hat{U}^{\circ}_r + PV)$ $T = 25^{\circ}\text{C} = 298.15\text{ K}$

$$\Delta \hat{H}^{\circ}_r = \Delta \hat{U}^{\circ}_r + \Delta n_g RT$$

$$\Delta n_p = (n_p)_{\text{product}} - (n_p)_{\text{reactants}} \text{ for gas only}$$

$$2 + 5 = 7 \text{ moles}$$

$$R = 8.314 \text{ J/Kmol}$$

$$\Delta \hat{U}^{\circ}_r = \Delta \hat{H}^{\circ}_r - \Delta n_g RT$$

$$= 69360 \text{ J} - (7 \text{ mol} \times 8.314 \text{ J/Kmol} \times 298.15 \text{ K})$$

$$= 52008 \text{ J} = \boxed{52.008 \text{ KJ}} = \Delta \hat{U}^{\circ}_r$$

→ physical significance

= $\Delta \hat{U}^{\circ}_r$ tells us that the total internal energy of the product is greater than the reactants

c) $\text{CaC}_2 = 150.0 \text{ g}$ rigid container at 25°C

$$M_r \text{ CaC}_2 = 64.1 \text{ g/mol}$$

↳ means constant V

$$\text{moles} = \frac{150}{64.1} = 2.34 \text{ mol CaC}_2$$

$$\Delta \hat{U}^{\circ}_r = Q_p$$

$$Q_p = \Delta(\hat{U}^{\circ}_r + PV)$$

$$Q_p = 52.008 \text{ KJ for 1 mole}$$

$$2.34 \text{ moles} (52.008 \text{ KJ/mol}) = 121.699 \text{ KJ}$$

→ the direction of heat transfer is from the reactor with an amount of 121.699 KJ , the "t" value for heat

bomb
calorimeter

(2)

• volume is
constant



$$\Delta U + \Delta KE^0 + \Delta PE^0 = Q + W$$

For water

$$U_{sys,F} - U_{sys,0} = Q$$

$$Q = m C_v (T_F - T_0)$$

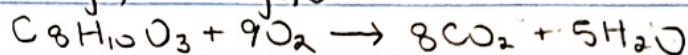
$$C_p = C_v = 4.18 \text{ J/g}^\circ\text{C}$$

$$= 650(4.18 \text{ J/g}^\circ\text{C})(36.2 - 23.6^\circ\text{C})$$

$$= 34234 \text{ J}$$

* For chamber

$$U_{sys,F} - U_{sys,0} = Q = -34234 \text{ J}$$



$$\bullet \text{ moles} = \underline{2.75} = 0.0179 \text{ g/mol}$$

$$\bullet U_{sys,F} - U_{sys,0} = Q = n_r \Delta \hat{U}^0_r = -34234 \text{ J}$$

(change from moles in gas phase $Q - S = -1 \text{ mol}$)

$$\bullet n_r (\Delta \hat{U}^0_r) = 0.0179 \text{ mol} - 34234$$

$$\Delta \hat{U}^0_r = -1956229$$

$$= -1956229 = -1956.229 \text{ kJ/mol}$$

$$\bullet \Delta \hat{H}^0_r = \Delta \hat{U}^0_r + RT \Delta n_R$$

$$= -1956 + (0.0083144 \times 298 \times (-1))$$

$$\Delta \hat{H}^0_r = -1958 \text{ kJ/mol}$$

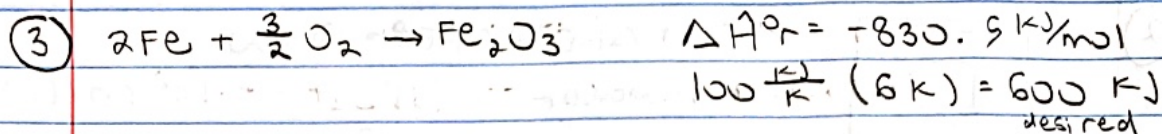
$$\bullet \Delta \hat{H}^0_r = \sum v_i \Delta \hat{H}_{F,i}$$

$$= (-1) \Delta \hat{H}_F$$

$$= \text{C}_8\text{H}_{10}\text{O}_3 + (-9)(0) + 8(-393.5) + 5(-285.84) = 1958$$

$$+ 0 - 3148 - 1429.2 = 1958$$

$$\Delta \hat{H}^0_r = 2592.2 \text{ kJ/mol}$$



Scale function $\frac{-600}{-830.5} = 0.7224 \text{ gmol Fe}_2\text{O}_3$

$2 \cdot 0.7224 \text{ Fe} = 1.44 \text{ gmol Fe}$

$\text{Mr Fe} = 56 \text{ g/gmol}$

• $\text{Fe in pack} = 56 \frac{\text{g}}{\text{mol Fe}} (1.44 \text{ gmol Fe})$

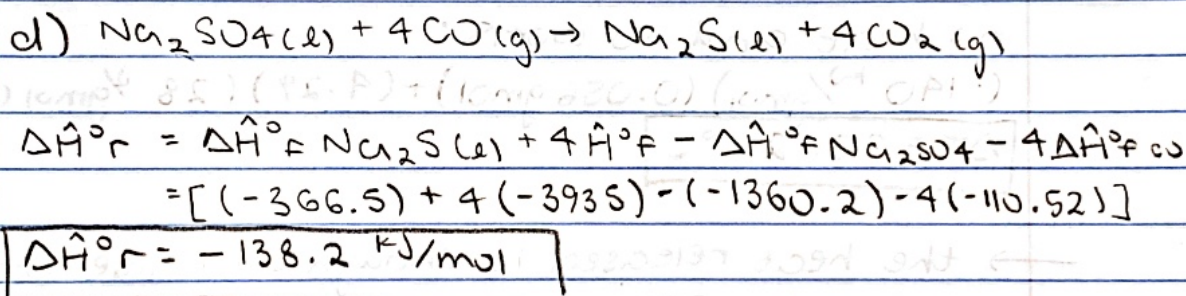
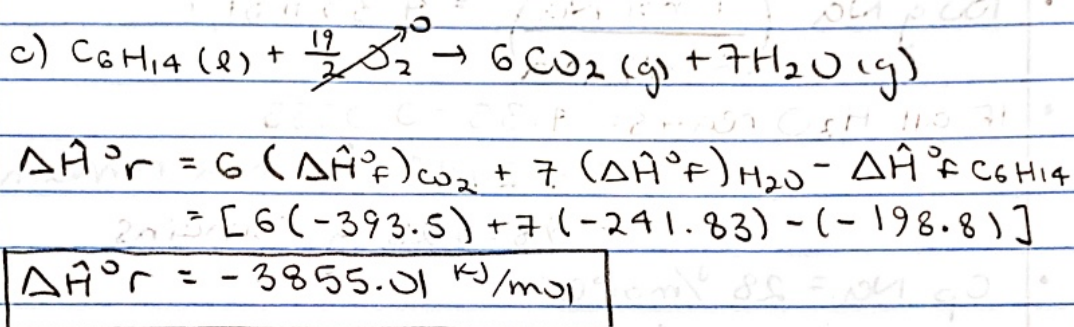
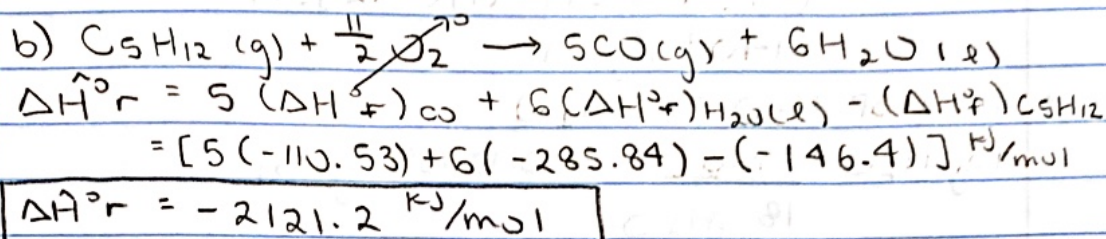
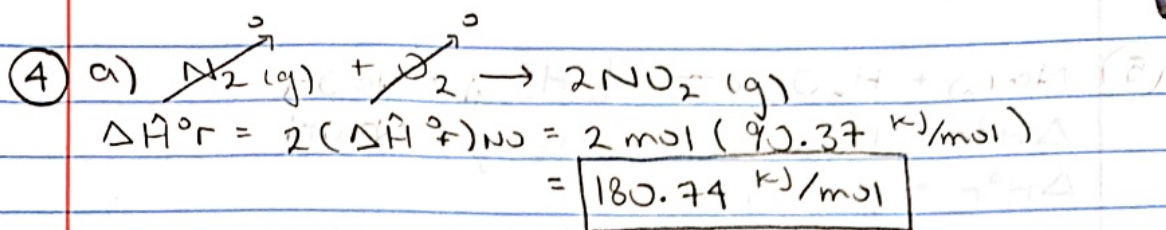
$\text{Fe in pack} = 80.64 \text{ g}$

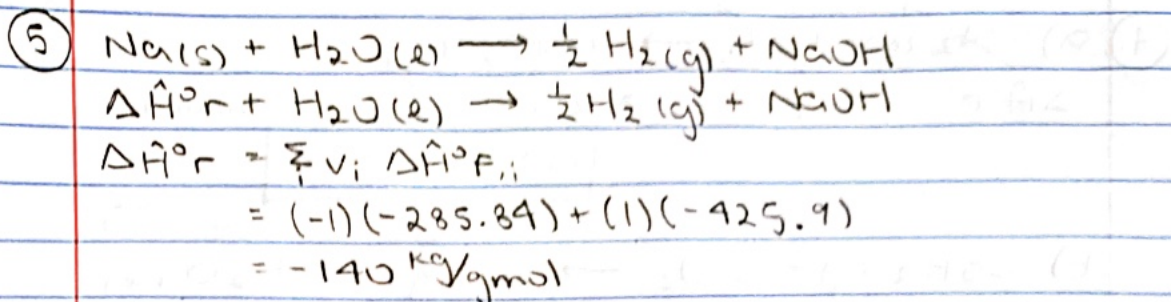
$\text{Mr Fe}_2\text{O}_3 = 160 \text{ g/gmol}$

• $0.7224 (160) = 115.584 \text{ g Fe (after use)}$

→ $115.584 - 80.64 = 34.944 \text{ g difference}$

• $\frac{115.584 - 80.64}{80.64} \times 100\% = 43.33\% \text{ difference}$





- 1 ml of $\text{H}_2\text{O} = 1 \text{ g H}_2\text{O}$

$$1 \text{ g H}_2\text{O} = \frac{1 \text{ mol}}{18 \text{ g H}_2\text{O}} = 0.0555 \text{ gmol}$$

- 100 g Na $\left(\frac{1 \text{ mol Na}}{22.99 \text{ g Na}} \right) = 4.35 \text{ mol}$

- IF all H_2O reacts: $4.35 - 0.0555$
 $= 4.2944 \text{ mol Na remains}$
 $= 98.728 \text{ g remains}$

- $C_p \text{ Na} = 28 \text{ J/mol}^\circ\text{C}$

no heat and no work:

$$(-140 \text{ kJ/gmol})(0.056 \text{ gmol}) + (4.29)(28 \text{ J/gmol}^\circ\text{C})(\Delta T) = 0$$

$$\Delta T = 65.27^\circ\text{C}$$

- the heat released is enough to raise the sodium by 65°C , which would raise the temperature 40°C from room temperature (25°C). But, Na can react violently and cause the hydrogen to explode.
- there is enough water to absorb the heat of reaction, which would raise the temperature but would not ignite H_2 .